NATIONAL CADET CORPS

NCC MANAGEMENT SYSTEM

PROJECT DOCUMENTATION

External Examination — Complete Project Explanation

Project Title	NCC Management System — Online Portal
Student Name	Sakshi Sahani
Technology Used	HTML, CSS, Bootstrap, JavaScript, Node.js, MySQL
Project Type	Full Stack Web Application
Year	2026
Purpose	Digital NCC cadet enrollment & management

1. Project Overview

The NCC Management System is a full-stack web application designed to digitize the entire process of NCC cadet enrollment, management and information access. The system replaces manual paper-based processes with an efficient, secure and user-friendly online portal.

1.1 PROBLEM STATEMENT

Before this system, NCC units faced the following challenges:

- Manual paper-based cadet registration — error-prone and time-consuming
- No centralized database for cadet information
- Difficulty accessing study materials and handbooks
- No easy way for admin to manage, search and export cadet data
- Physical documents prone to loss and damage

1.2 SOLUTION PROVIDED

This project provides:

- Online cadet registration with validation
- Secure login system with password encryption
- Complete cadet detail form with 30+ fields
- Admin dashboard to view, search, edit and delete cadet records
- Wing-wise study material and handbooks accessible online
- Export to CSV and Excel for reporting
- PDF generation of cadet enrollment form

2. Technologies Used

Layer	Technology	Purpose
Frontend	HTML5	Structure of all web pages
Frontend	CSS3 + Bootstrap 5	Styling, responsive design, components
Frontend	JavaScript (Vanilla)	Form validation, dynamic behaviour, API calls
Frontend	Font Awesome 6	Icons throughout the interface
Frontend	Google Fonts (Rajdhani + Barlow)	Typography for NCC theme
Backend	Node.js	Server-side JavaScript runtime
Backend	Express.js	Web framework — handles API routes
Backend	bcrypt	Password hashing and security
Backend	cors	Cross-Origin Resource Sharing
Database	MySQL (mysql2)	Relational database for all data
PDF	jsPDF	Client-side PDF generation
Auth	Session Storage	Admin authentication state

3. Project Modules

The project is divided into the following major modules:

3.1 Public Website (Frontend)

Page	File	Description
Home Page	index.html	Landing page with carousel, wings section, footer, news ticker
Army Wing	army-wing.html	Army wing resources — study material, handbooks, jobs, camps
Navy Wing	navy-wing.html	Navy wing resources — seamanship, navigation, jobs
Air Wing	air-wing.html	Air wing resources — aviation, aero-modelling, jobs
Army Syllabus	army-syllabus.html	8-chapter study notes — NCC intro to leadership
Army Drill	army-drill-handbook.html	5-chapter drill manual — positions to arms drill
Navy Syllabus	navy-syllabus.html	7-chapter notes — seamanship, knots, navigation
Air Syllabus	air-syllabus.html	7-chapter notes — flight principles, aero-modelling

3.2 NCC Cadet Registration System

File: ncc-form.html

This is a 3-step multi-page form:

- Step 1 — Registration: Cadet creates account with ID, name, phone, BN Code, password
- Step 2 — Login: Cadet logs in with ID + BN Code + Password
- Step 3 — Cadet Details: Full profile with photo, personal, academic and NCC information

Fields in Cadet Details Form:

Section	Fields
Photo & Identity	Passport photo upload, BN Code, Mobile, Unit Name, Group HQ
Personal Info	DOB, Gender, Blood Group, Category, Religion, Nationality, Address, Father Name, Emergency Contact
Academic Info	School/College, Class/Year, Stream

Section	Fields
NCC Unit Details	Wing, Rank, Enrollment Year, Certificate, Directorate, Training Status, Camps, Drill Proficiency, Awards, Sports
Documents	Aadhar, Marksheet, Medical, NCC Certificate, Character Certificate, Domicile

3.3 Admin Dashboard

File: dashboard.html

- Secure admin login — username + password protected
- Statistics cards — Total Cadets, With Details, Army Wing, Today's Registrations
- Full data table with all cadet information — 14 columns
- Search by name, Cadet ID, BN Code, phone, school, father name
- Filter by Wing and Gender
- Sort by any column (ascending / descending)
- Edit cadet — modal with all fields editable
- Delete cadet — confirmation modal
- Pagination — 10 records per page
- Export to CSV — all fields
- Export to Excel — formatted spreadsheet

3.4 Backend API (server.js)

Method	Route	Function
POST	/api/register	Register new cadet — validates + hashes password + saves to DB
POST	/api/login	Cadet login — verify ID, BN Code, password
POST	/api/cadet-details	Save full cadet details — all 30+ fields
GET	/api/cadets	Get all cadets — JOIN query for dashboard
GET	/api/cadets/:id	Get single cadet by ID
PUT	/api/cadets/:id	Update cadet — registration + detail fields
DELETE	/api/cadets/:id	Delete cadet — cascades to details
POST	/api/admin/login	Admin authentication

4. Database Design

Database: MySQL | Database Name: ncc_system

4.1 Table: cadets

Column	Type	Description
id	INT AUTO_INCREMENT PK	Unique row identifier
cadet_id	VARCHAR(20) UNIQUE	Primary key used for login (e.g. NCC2024001)
first_name	VARCHAR(60) NOT NULL	First name
middle_name	VARCHAR(60)	Middle name (optional)
last_name	VARCHAR(60) NOT NULL	Last name
phone_number	VARCHAR(15) NOT NULL	Registered phone
bn_code	VARCHAR(20) NOT NULL	Battalion code
password	VARCHAR(255) NOT NULL	bcrypt hashed password
last_login	DATETIME	Last login timestamp
created_at	DATETIME	Registration timestamp

4.2 Table: cadet_details

Column	Type	Description
cadet_id	VARCHAR(20) UNIQUE FK	Foreign key → cadets.cadet_id
mobile	VARCHAR(15)	Detail form mobile number
unit_name	VARCHAR(100)	NCC unit name
dob	DATE	Date of birth
gender	VARCHAR(10)	Male / Female / Other
blood_group	VARCHAR(5)	A+, B-, O+ etc.
address	TEXT	Permanent address
father_name	VARCHAR(100)	Father / Guardian name
emergency_phone	VARCHAR(15)	Emergency contact
school_name	VARCHAR(150)	School or college
wing	VARCHAR(30)	Army / Navy / Air Force Wing
rank	VARCHAR(30)	Cadet rank
enrollment_year	YEAR	Year of enrollment
certificate	VARCHAR(20)	A / B / C Certificate
directorate	VARCHAR(80)	State NCC Directorate

Column	Type	Description
training_status	VARCHAR(50)	Current training status
awards	VARCHAR(200)	NCC awards and medals
saved_at	DATETIME	Last saved timestamp

Relationship: cadets (1) ←→ (1) cadet_details — One-to-One with CASCADE DELETE

5. System Flow Diagram

Cadet Flow:

1. Cadet visits index.html → clicks 'Join Now' or 'NCC-Form'
2. Fills Registration form (Step 1) → data saved to cadets table
3. Success overlay shows Cadet ID → cadet notes it down
4. Logs in (Step 2) → BN Code + Cadet ID + Password verified
5. Fills Cadet Details (Step 3) → data saved to cadet_details table
6. Success overlay → cadet can Download PDF of enrollment form

Admin Flow:

7. Admin visits admin-login.html → enters username + password
8. Authenticated → redirected to dashboard.html
9. Views all cadets in table → can search, filter, sort
10. Can Edit any cadet record via Edit modal
11. Can Delete cadet (with confirmation)
12. Can Export all data as CSV or Excel
13. Logout → session cleared → back to login

Cadet Wing Resources Flow:

14. Cadet visits index.html → clicks Learn More on any wing card
15. Opens wing page (army/navy/air-wing.html)
16. Sees study material list → clicks Download → syllabus page opens
17. Sees handbook list → clicks Read Online → handbook page opens
18. All content organized in chapter tabs — can print/save as PDF

6. Security Features

Feature	How it works
Password Hashing	bcrypt library hashes password with salt rounds=10 before storing in DB — plain password never saved
Duplicate Check	Cadet ID and phone number checked for duplicates during registration
Input Validation	Both frontend (JS) and backend (Node.js) validate all required fields
Admin Auth	sessionStorage stores adminLoggedIn + token — dashboard auto-redirects if not logged in
SQL Injection Prevention	Parameterized queries used in all MySQL execute() calls — no string concatenation
Age Validation	DOB validated on server — age must be 13-26 years
Password Strength	Frontend shows live password strength indicator (Weak/Fair/Good/Strong)
Photo Validation	Client-side check — only JPG/PNG under 2MB accepted

7. Complete Features List

7.1 Frontend Features

- Responsive design — works on desktop, tablet and mobile
- Bootstrap 5 grid system for layout
- Sticky navbar with active page highlighting
- CSS animation news ticker on home page
- Auto-sliding image carousel on home page (carousel-fade)
- Animated floating particles background on form
- Progress bar showing form completion (33% → 66% → 100%)
- Live password strength indicator
- Photo upload with live preview
- Document upload with visual confirmation
- Success overlays with animations after form submission
- Login progress bar animation before redirecting
- Chapter-tab navigation in study material pages
- Print / Save as PDF button on all content pages

7.2 Backend Features

- RESTful API with 8 endpoints
- MySQL connection pool — handles multiple users
- Auto table creation and migration on server start
- ON DUPLICATE KEY UPDATE — upsert for cadet details
- FOREIGN KEY with CASCADE DELETE — data integrity
- Async/await throughout — no callback hell
- Error handling with meaningful messages returned to frontend
- Console logging for all important events

7.3 Dashboard Features

- 4 statistics cards with live counts
- Real-time search across 5 fields
- Wing filter and Gender filter
- Column sorting (click header to sort asc/desc)
- Pagination — 10 records per page with navigation buttons
- Edit modal — all fields editable including extended fields
- Delete confirmation modal — prevents accidental deletion
- Toast notifications for all actions
- Loading spinner while fetching data
- Export CSV — 25+ fields
- Export Excel — formatted HTML table

8. Project File Structure

File	Type	Description
index.html	HTML	Home page — carousel, wings section, footer
ncc-form.html	HTML + JS	Registration → Login → Cadet Details (3-step form)
dashboard.html	HTML + JS	Admin dashboard — view/edit/delete/export cadets
admin-login.html	HTML + JS	Admin login page
army-wing.html	HTML	Army wing page — study material + handbooks
navy-wing.html	HTML	Navy wing page — study material + handbooks
air-wing.html	HTML	Air wing page — study material + handbooks
army-syllabus.html	HTML	Army syllabus — 8 chapters NCC content
army-drill-handbook.html	HTML	Army drill manual — 5 chapters
navy-syllabus.html	HTML	Navy syllabus — 7 chapters NCC content
air-syllabus.html	HTML	Air wing syllabus — 7 chapters NCC content
server.js	Node.js	Backend server — all 8 API routes + DB setup
ncc.db / ncc_system	MySQL DB	Two tables: cadets + cadet_details
images/	Folder	logo.jpg, army.jpg, navy.jpg, airforce.jpg, ncc1-3.jpg
package.json	Config	Node dependencies — express, mysql2, bcrypt, cors

9. How to Run the Project

STEP 1 — PREREQUISITES

- Node.js installed (v16 or above)
- MySQL installed and running
- Any web browser (Chrome recommended)

STEP 2 — SETUP

19. Open terminal in project folder
20. Run: npm install express mysql2 bcrypt cors
21. Open MySQL and create database: CREATE DATABASE ncc_system;
22. Edit server.js — set your MySQL password in dbConfig

STEP 3 — START SERVER

23. Run: node server.js
24. Server starts on http://localhost:3000
25. Tables are auto-created on first run

STEP 4 — ACCESS

- Open browser → http://localhost:3000 → Home page opens
- Click NCC-Form → Registration → Login → Cadet Details
- Admin Dashboard → http://localhost:3000/dashboard.html
- Admin Credentials: username: admin | password: ncc@admin123

10. Expected External Questions & Answers

Q1. What is the purpose of this project?

This project digitizes the NCC cadet enrollment and management process. It allows cadets to register online, submit their complete details and access study materials. The admin can manage all cadet records from a central dashboard.

Q2. What technology stack did you use?

Frontend: HTML, CSS, Bootstrap 5, JavaScript. Backend: Node.js with Express.js framework. Database: MySQL with mysql2 library. Security: bcrypt for password hashing. PDF: jsPDF library for client-side PDF generation.

Q3. How is the password stored securely?

The password is never stored in plain text. We use the bcrypt library which applies a hashing algorithm with a salt (random value) and 10 rounds. Even if the database is compromised, the actual password cannot be recovered.

Q4. Explain the database design.

There are two tables: (1) cadets — stores registration data like name, phone, BN Code and hashed password. (2) cadet_details — stores full profile linked by cadet_id as foreign key. They have a One-to-One relationship. CASCADE DELETE ensures that if a cadet is deleted, their details are also deleted automatically.

Q5. What validations have you implemented?

Frontend validations: required field checks, 10-digit Indian mobile validation, password strength indicator, age validation (13-26 years), photo size/type check. Backend validations: all the same checks repeated in server.js so they cannot be bypassed by disabling JavaScript.

Q6. How does the admin dashboard work?

The dashboard fetches all cadet data using a GET /api/cadets API call which does a LEFT JOIN between cadets and cadet_details tables. Data is displayed in a paginated table. Admin can search, filter by wing/gender, sort by columns, edit records via a modal form, delete with confirmation, and export to CSV or Excel.

Q7. What is a REST API?

REST (Representational State Transfer) API is a way for frontend and backend to communicate using HTTP methods: GET (read data), POST (create data), PUT (update data), DELETE (remove data). Our server.js has 8 REST API endpoints that the HTML pages call using the fetch() function.

Q8. How does the cadet registration process work?

Step 1: Cadet fills registration form and clicks Register. JavaScript validates the form. A POST request is sent to /api/register. Server checks for duplicate Cadet ID and phone. Password is hashed with bcrypt. Record is saved to cadets table. A success modal shows the Cadet ID. Step 2: Cadet logs in. Step 3: Cadet fills complete details form. All 30+ fields are saved to cadet_details table.

Q9. What is Bootstrap and why did you use it?

Bootstrap is a CSS framework that provides ready-made responsive components like navbar, cards, grid system, forms and buttons. We used Bootstrap 5 to make the site responsive (works on mobile and desktop) and to save time on basic styling while focusing on custom NCC-themed CSS.

Q10. What are the limitations of your project?

- No real file upload server — photo and documents are shown locally but not saved to server
- Admin password is hardcoded — should be stored in database with hashing
- No email verification for cadet registration
- No forgot password feature
- Single admin account — no role-based access
- No SSL/HTTPS — for production deployment, HTTPS is required

11. Conclusion

The NCC Management System successfully achieves its goal of digitizing the NCC cadet enrollment and management process. The project demonstrates full-stack web development skills — from frontend design with HTML/CSS/Bootstrap to backend API development with Node.js and Express, to database design with MySQL.

The system provides value to NCC units by eliminating paperwork, maintaining a searchable database of cadets, providing online study resources and giving administrators a powerful dashboard to manage cadet information efficiently.

Future enhancements could include: actual file storage using Multer, email notifications, SMS alerts for camp notifications, a mobile app, and a cadet performance tracking module.

End of Project Documentation

Developed by: Sakshi Sahani | NCC Management System | 2026